LAB:01 DATE:26|02|2026

Q**1.** Install the JDK on your computers and check the Java Compiler version and Java Interpreter Version installed.

```
gauravmathur — -zsh — 80×24

Last login: Thu Mar 12 09:22:13 on ttys000
gauravmathur@Annus-Mac ~ % java -version
java version "19.0.1" 2022-10-18
Java(TM) SE Runtime Environment (build 19.0.1+10-21)
Java HotSpot(TM) 64-Bit Server VM (build 19.0.1+10-21, mixed mode, sharing)
gauravmathur@Annus-Mac ~ % javac -version
javac 19.0.1
gauravmathur@Annus-Mac ~ %
```

Q:2 Create a Java source file named **`MultiClassDemo.java`** that contains **three classes**:

- Class `First`
- Class `Second`
- Class `MultiClassDemo`

Each class must contain its **own `main()` method**.

Each `main()` method should print a different message identifying the class being executed.

Observe how many `.class` files are generated.

MultiClassDemo.java > Second

```java

public class MultiClassDemo {
    Run | Debug
    public static void main(String[] args) {
        System.out.print(s: "This is MultiClassDemo class");
    }
}
class First{
    Run | Debug
    public static void main(String[] args) {
        System.out.println(x: "Hey i am first class");
    }
}
class Second{
    Run | Debug
    public static void main(String[] args) {
        System.out.println(x: "Hey i am Second class");
    }
}
```

Output:-

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
* History restored

gauravmathur@Annus-Mac LeetCode % java MultiClassDemo
This is MultiClassDemo class%
gauravmathur@Annus-Mac LeetCode %
```

Q3:Create a program named `MyFirstProgram.java` to print:

- Your name
- Roll number
- Department
- Email

MyFirstProgram.java > ...

```java
public class MyFirstProgram {
    Run | Debug
    public static void main(String[] args) {
        System.out.println(x: "My name is Annu");
        System.out.println(x: "My roll Number is : 2023BCSE008");
        System.out.println(x: "My Department is CSE");
        System.out.println(x: "my email is : annumathur003@gmail.com");
    }

}
```

Output:—

```
^C
gauravmathur@Annus-Mac LeetCode %  cd /Users/gauravmathur/Desktop/LeetCode ; /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-19.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/gauravmathur/Library/Application\ Support/Code/User/workspaceStorage/e61c234f5c98a194c51783627f3b6cfc/redhat.java/jdt_ws/LeetCode_ad607690/bin MyFirstProgram
My name is Annu
My roll Number is : 2023BCSE008
My Department is CSE
my email is : annumathur003@gmail.com
gauravmathur@Annus-Mac LeetCode %
```

Q4:Create a program to take the following as command line inputs from the user and printing them:

- Your name
- Roll number
- Department
- Email

SecondProgram.java > ...

```java
import java.util.Scanner;
public class SecondProgram {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter your name:");
        String name = sc.nextLine();
        System.out.println(x: "Enter your roll no:");
        String roll = sc.nextLine();
        System.out.println(x: "Enter your Department");
        String dept = sc.nextLine();
        System.out.println(x: "Enter your email");
        String email = sc.nextLine();
        System.out.println("Your name is :"+name);
        System.out.println("Your roll number is :"+roll);
        System.out.println("Your department is :"+dept);
        System.out.println("Your email is :"+email);
        sc.close();



    }


}

```

Output:—-

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
r/Library/Application\ Support/Code/User/workspaceStorage/e61c234f5c98a194c51783627f3b6cfc/redhat.java/jdt_ws/
LeetCode_ad607690/bin SecondProgram
^C%
gauravmathur@Annus-Mac LeetCode %  /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-19.jdk/Contents/Home/bin
/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/gauravmathur/Library/Application\ Support/Code/User/w
orkspaceStorage/e61c234f5c98a194c51783627f3b6cfc/redhat.java/jdt_ws/LeetCode_ad607690/bin SecondProgram
Enter your name:
Annu
Enter your roll no:
2023bcse008
Enter your Department
cse
Enter your email
annumathur003@gmail.com
Your name is :Annu
Your roll number is :2023bcse008
Your department is :cse
Your email is :annumathur003@gmail.com
gauravmathur@Annus-Mac LeetCode %
```

Run:... zsh Run:... Run:...

Ln 25, Col 1 Spaces: 4 UTF-8 LF { } Java Go Live

Q5. Write a Java program that declares variables of all primitive data types and prints their values. Use:

byte

short

int

long

float

double

char

boolean

PrimitiveDemo.java > PrimitiveDemo

```java
public class PrimitiveDemo {
    Run | Debug
    public static void main(String[] args) {
        byte b = 10;
        short s = 200;
        int i = 5000;
        long l = 100000L;
        float f = 5.5f;
        double d = 99.99;
        char c = 'A';
        boolean bool = true;
        System.out.println("Byte value: " + b);
        System.out.println("Short value: " + s);
        System.out.println("Int value: " + i);
        System.out.println("Long value: " + l);
        System.out.println("Float value: " + f);
        System.out.println("Double value: " + d);
        System.out.println("Char value: " + c);
        System.out.println("Boolean value:"+bool);
    }
}
```

Output:—

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
r/Library/Application\ Support/Code/User/workspaceStorage/e61c234f5c98a194c51783627f3b6cfc/redhat.java/jdt_ws/
LeetCode_ad607690/bin PrimitiveDemo
gauravmathur@Annus-Mac LeetCode %  /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-19.jdk/Contents/Home/bin
/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/gauravmathur/Library/Application\ Support/Code/User/w
orkspaceStorage/e61c234f5c98a194c51783627f3b6cfc/redhat.java/jdt_ws/LeetCode_ad607690/bin PrimitiveDemo
Byte value: 10
Short value: 200
Int value: 5000
Long value: 100000
Float value: 5.5
Double value: 99.99
Char value: A
Boolean value:true
gauravmathur@Annus-Mac LeetCode %
```

Run:... zsh Run:... Run:... Run:...

Ln 20, Col 2 Spaces: 4 UTF-8 LF { } Java Go Live

Q6: Write a program that:

Takes two integers (hardcoded)

Performs:

Addition

Subtraction

Multiplication

Division

Modulus

Prints all results.

Program6.java > ...

```java
public class Program6 {
    Run | Debug
    public static void main(String[] args) {
        int a = 20;
        int b = 5;
        System.out.println(a+b);
        System.out.println(a-b);
        System.out.println(a*b);
        System.out.println(a/b);
        System.out.println(a%b);
    }

}
```

Output:-

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
gauravmathur@Annus-Mac LeetCode %  cd /Users/gauravmathur/Desktop/LeetCode ; /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-19.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/gauravmathur/Library/Application\ Support/Code/User/workspaceStorage/e61c234f5c98a194c51783627f3b6cfc/redhat.java/jdt_ws/LeetCode_ad607690/bin Program6
0
gauravmathur@Annus-Mac LeetCode %  /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-19.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/gauravmathur/Library/Application\ Support/Code/User/workspaceStorage/e61c234f5c98a194c51783627f3b6cfc/redhat.java/jdt_ws/LeetCode_ad607690/bin Program6
25
15
100
4
0
gauravmathur@Annus-Mac LeetCode %
```

Run:... zsh Run:... Run:... Run:... Run:...

Ln 13, Col 1 Spaces: 4 UTF-8 LF { } Java Go Live

Q7:Write a program that:

Declares two integers and

Performs:

&

|

^

~

Program7.java > Program7 > main(String[])

```java
public class Program7 {
    Run | Debug
    public static void main(String[] args) {
        int a = 5;
        int b = 4;
        System.out.println(x: "&(AND):");
        System.out.print(a&b);
        System.out.print(s: "|(or):");
        System.out.println(a|b);
        System.out.print(s: "^(XOR):");
        System.out.println(a^b);
        System.out.print(s: "~:");
        System.out.println(~a);
        System.out.print(~b);
    }

}
```

Output:—

```
gauravmathur@Annus-Mac LeetCode %  cd /Users/gauravmathur/Desktop/LeetCode ; /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-19.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/gauravmathur/Library/Application\ Support/Code/User/workspaceStorage/e61c234f5c98a194c51783627f3b6cfc/redhat.java/jdt_ws/LeetCode_ad607690/bin Program7
&(AND):
4|(or):5
^(XOR):1
~:-6
-5%
gauravmathur@Annus-Mac LeetCode %
```

Ln 11, Col 32 Spaces: 4 UTF-8 LF { } Java

LAB:02 DATE: 12|03|2026

Q1. WAP using functions (methods) to find the sum of two numbers.

first.java > first > main(String[])

```java
import java.util.Scanner;
public class first{
    public static int sum(int a,int b){
        int  sum = a+b;
        System.out.println("The sum is :"+sum);
        return sum;
    }
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the first number:");
        int a = sc.nextInt();
        System.out.println(x: "Enter second number:");
        int b = sc.nextInt();
        sum(a,b);
        sc.close();

    }
}
```

OUTPUT:-

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
gauravmathur@Annus-Mac LeetCode % java first
Enter the first number:
4
Enter second number:
8
The sum is :12
gauravmathur@Annus-Mac LeetCode %
```

Q2: WAP to calculate the remainder of dividing two numbers and print the output as "42 divided by 2 gives remainder 0".

lab_02_02.java > ...

```java
import java.util.Scanner;
public class lab_02_02 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the first number:");
        int a = sc.nextInt();
        System.out.println(x: "Enter second number:");
        int b = sc.nextInt();
        System.out.println(a +" divided by "+ b +" gives remainder :"+(a%b));
        sc.close();

    }

}
```

OUTPUT:

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
gauravmathur@Annus-Mac LeetCode % java lab_02_02.java
Enter the first number:
4
Enter second number:
8
4 divided by 8 gives remainder :4
gauravmathur@Annus-Mac LeetCode %
```

Q3:Write a Java program to calculate the area and perimeter of a rectangle using arithmetic operators.

lab_02_03.java > lab_02_03 > main(String[])

```java
import java.util.Scanner;
public class lab_02_03 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the length of rectangle:");
        int a = sc.nextInt();
        System.out.println(x: "Enter the breadth of rectangle:");
        int b = sc.nextInt();
        System.out.println("Area is:"+(a*b));
        System.out.println("perimeter is"+(2*(a+b)));
        sc.close();

    }

}
```

OUTPUT:-

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
gauravmathur@Annus-Mac LeetCode % java lab_02_03.java
Enter the length of rectangle:
4
Enter the breadth of rectangle:
8
Area is:32
perimeter is24
gauravmathur@Annus-Mac LeetCode %
```

Q4. Write a Java program to **calculate the average of three numbers**.

lab_02_04.java > lab_02_04 > main(String[])

```java
import java.util.Scanner;
public class lab_02_04 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the first number:");
        int a = sc.nextInt();
        System.out.println(x: "Enter second number:");
        int b = sc.nextInt();
        System.out.println(x: "Enter third number:");
        int c = sc.nextInt();
        System.out.println("The average of three numbers is :"+((a+b+c)/2));
        sc.close();


    }

}

```

OUTPUT:-

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
gauravmathur@Annus-Mac LeetCode % java lab_02_04.java
Enter the first number:
4
Enter second number:
8
Enter third number:
9
The average of three numbers is :10
gauravmathur@Annus-Mac LeetCode %
```

Ln 12, Co

5. WAP to find the square root of a number using predefined method and also using user-defined method of square_root (without predefined).

seven.java > SquareRoot > main(String[])

```java
import java.util.Scanner;

class SquareRoot {
    static int square_root(int n){
        int i = 1;
        while(i * i <= n){
            if(i * i == n)
                return i;
            i++;
        }
        return -1;
    }
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter number: ");
        int n = sc.nextInt();
        System.out.println("Using Math.sqrt(): " + Math.sqrt(n));
        System.out.println("Using user-defined method: " + square_root(n));
        sc.close();
    }
}
```

Output:—

PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
gauravmathur@Annus-Mac LeetCode % java Seven
Enter number: 16
Using Math.sqrt(): 4.0
Using user-defined method: 4
gauravmathur@Annus-Mac LeetCode %
```

Q6. Write a Java program that defines a method max taking two arguments to **returns the largest of two numbers**.

lab_02_08.java > lab_02_08 > main(String[])

```java
import java.util.Scanner;

public class lab_02_08 {
    public static void largest(int a,int b){
        if(a>b){
            System.out.println(x: "A is greater");
        }else{
            System.out.println(x: "B is greater");
        }
    }
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the first number:");
        int a = sc.nextInt();
        System.out.println(x: "Enter second number:");
        int b = sc.nextInt();
        largest(a, b);
        sc.close();

    }

}

```

OUTPUT:——

```
caused by: java.lang.ClassNotFoundException: lab_02_08
gauravmathur@Annus-Mac LeetCode % java lab_02_08.java
Enter the first number:
9
Enter second number:
2
A is greater
gauravmathur@Annus-Mac LeetCode %
```

Q7. WAP a program to find the factorial of a number using recursion (using methods)

lab_02_09.java > ...

```java
import java.util.Scanner;
public class lab_02_09 {
    public static int fact(int a){
        if(a==0 || a==1){
            return 1;
        }
        return a*fact(a-1);
    }
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter the  number:");
        int a = sc.nextInt();
        int res = fact(a);
        System.out.println("factorial is :"+res);
        sc.close();


    }

}

```

OUTPUT:—

PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
gauravmathur@Annus-Mac LeetCode % java lab_02_09.java
Enter the  number:
4
factorial is :24
gauravmathur@Annus-Mac LeetCode %
```